Basic Science Department

BAS 141
Engineering Chemistry

Ass.Prof. Nora Yehia Selem - Dr. Rasha Ahmed

Chapter 4: Solutions and its properties

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1.1. Introduction

The most common class of homogeneous mixtures, contains particles with diameters in the range of 0.1–2 nm, the size of a typical ion or small molecule. They don't separate on standing.



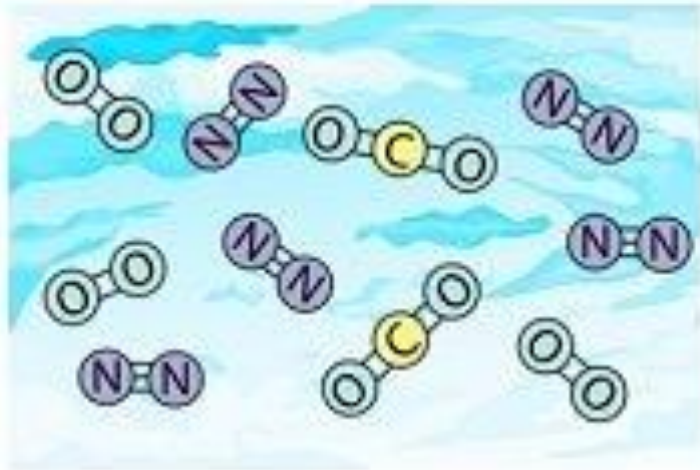
Kind of Solution	Example
Gas in gas	Air (O_2, N_2, Ar, and other gases)
Gas in liquid	Carbonated water (CO_2 in water)
Gas in solid	H_2 in palladium metal
Liquid in liquid	Gasoline (mixture of hydrocarbons)
Liquid in solid	Dental amalgam (mercury in silver)
Solid in liquid	Seawater (NaCl and other salts in water)
Solid in solid	Metal alloys, such as sterling silver (92.5% Ag, 7.5% Cu)

Common Examples of Solutions



Honey

It is a mixture of sugars and other carbohydrates



Air

It is a mixture containing gases nitrogen and oxygen with trace amount of other gases



Soda water

It is a mixture containing carbon-dioxide and water

Solid/Liquid Solutions



toothpaste



saltwater

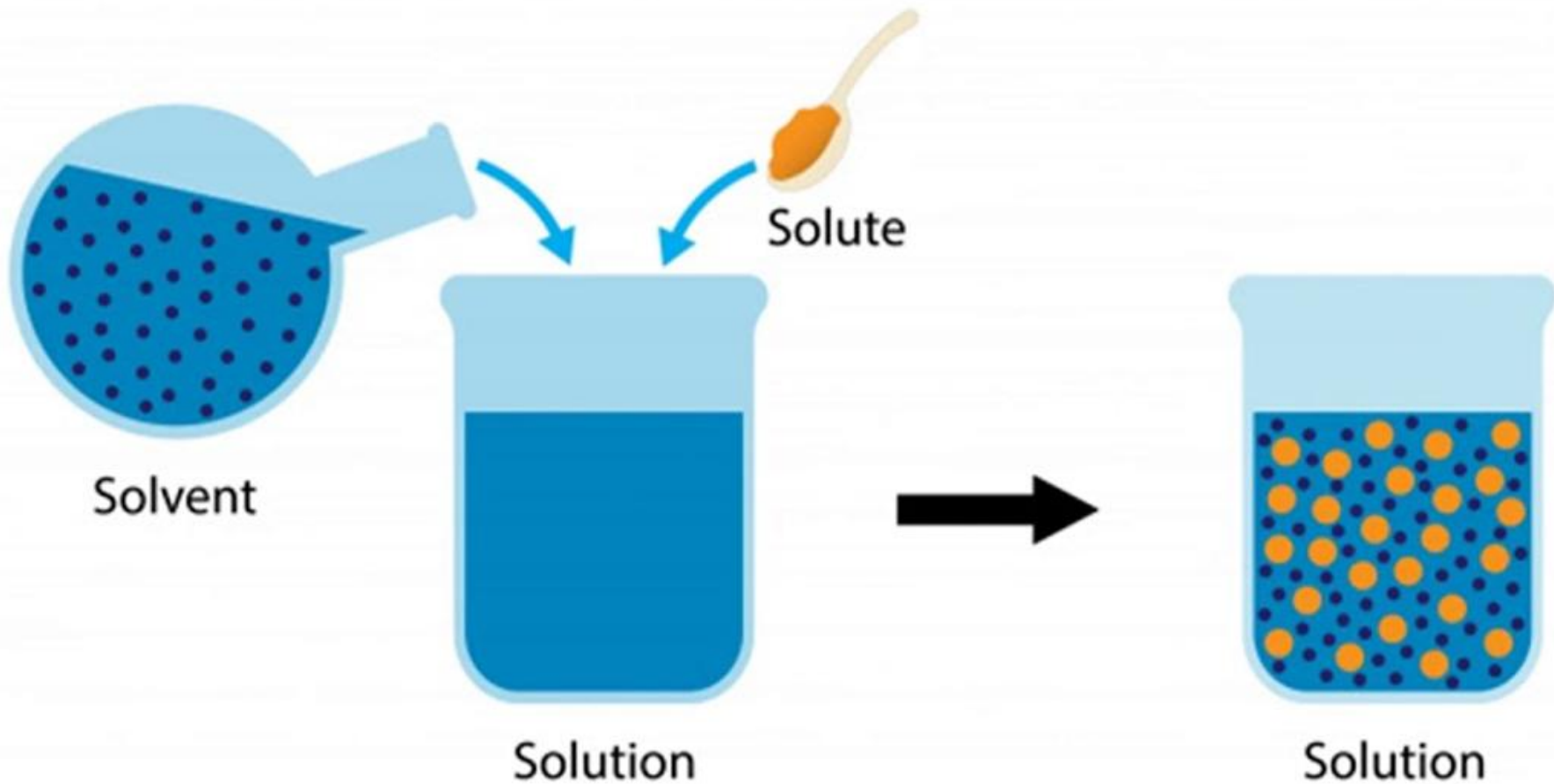


sugar water



sweetened tea or coffee

Solute + Solvent -> Solution



- The **solvent** is the material that usually decides the solution's physical state (solid, liquid, or gas).

- The **solute** is the product that the solvent dissolves.

- For solutions in which a **gas or solid is dissolved in a liquid**, the **dissolved substance** is called the **solute**, and the **liquid is called the solvent**.

- When one **liquid is dissolved in another**, the **minor component** is usually considered **the solute**, and the **major component is the solvent.**

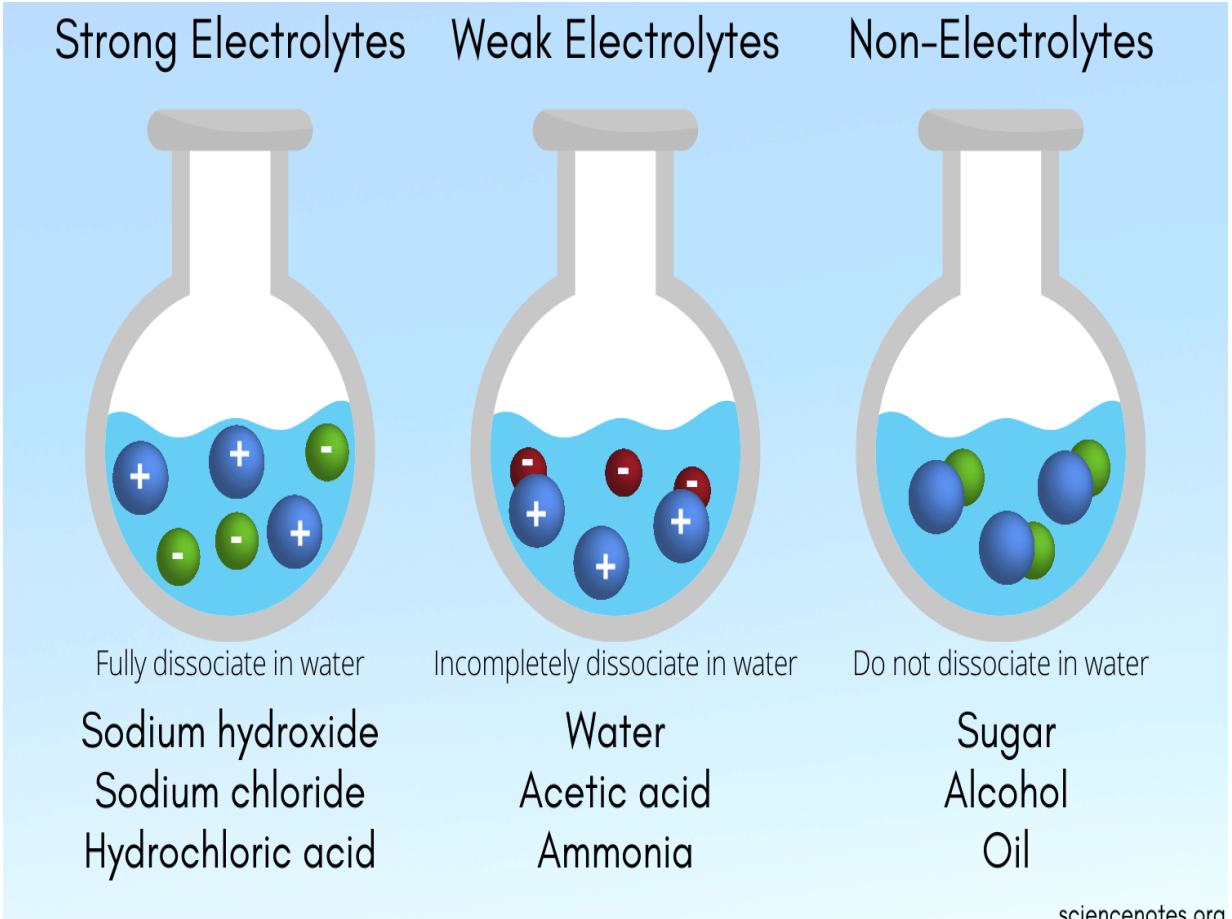
- An **aqueous solution** is a solution in which the **solvent** is water.

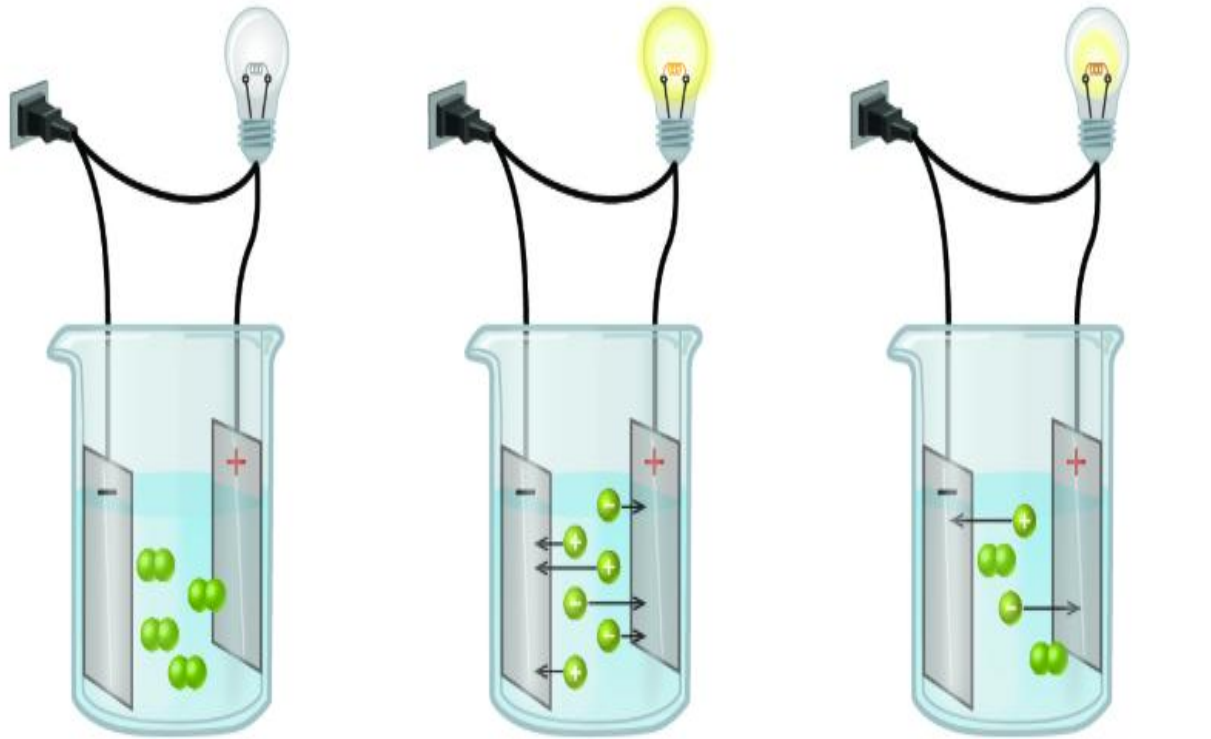
- **Alloys** are **solid solutions** as metal with **high concentration** is the **solvent** and the metal with **low concentration** is **the solute**.

1.2. Solution types

A. Based on electrical conductivity

Electrolytes solution conduct electricity.
Nonelectrolytes do not conduct electric current





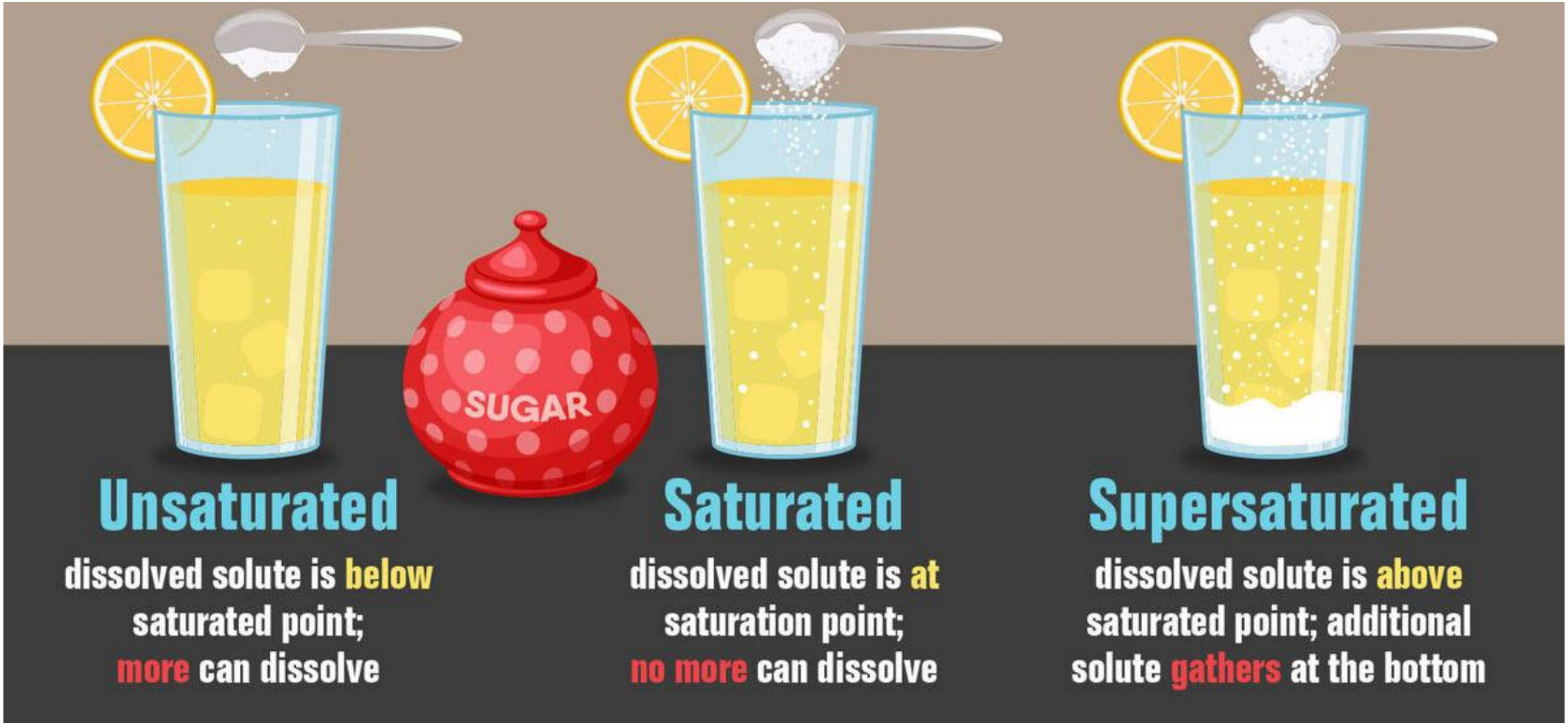
Solutions that are prepared using non-electrolytes do not contain ions and, therefore, do not conduct electricity.

Solutions that are prepared using strong electrolytes are primarily comprised of ions and, therefore, strongly conduct electricity.

Solutions that are prepared using weak electrolytes contain both ions and molecules and, therefore, weakly conduct electricity.

1.2. Solution types

B. Based on solubility

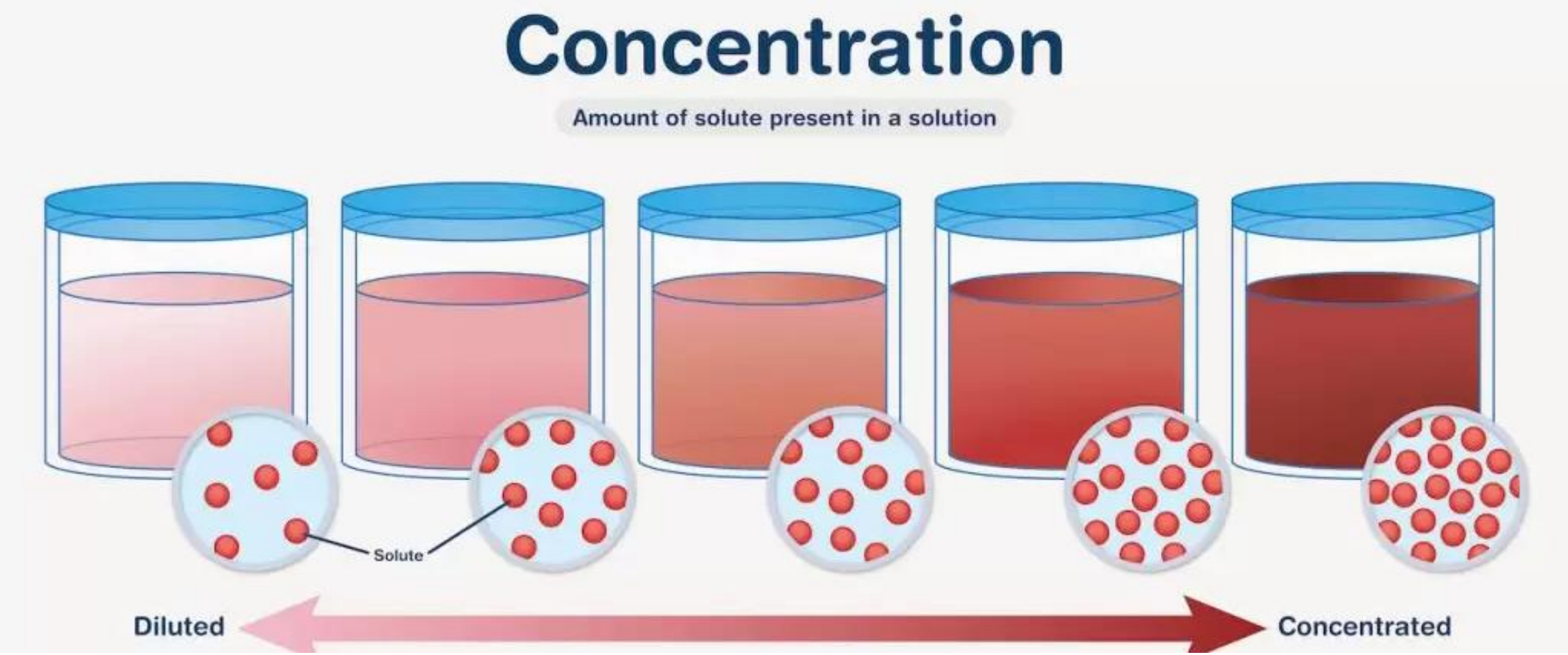


A saturated solution is a solution that contains the **maximum amount of solute** that is capable of dissolving in a solvent at a given temperature and pressure.

An unsaturated solution is a solution that contains **less than the maximum amount of solute** that is capable of being dissolved in a solvent at a given temperature and pressure.

A supersaturated solution is a solution that contains **more than the maximum amount** of solute that is capable of being dissolved at a given temperature.

1.3. Units of concentration



CONCENTRATION UNIT	SYMBOL	DEFINITION	DESCRIPTION
Molarity	M	$Mol.l^{-1}$	Moles per litre solution
Molality	none	$Mol.kg^{-1}$	Moles per kilogram solvent
Normality	N	$Val.l^{-1}$	Vales per litre solution
Weight per volume	none	$kg.l^{-1}$	Mass per volume
Percentage weight per volume	% (w/v)	-	Grams per 100 ml solution
Percentage weight per weight	% (w/w)	-	Grams per 100 g solution
Percentage volume per volume	% (v/v)	-	Millilitres per 100 ml solution

$$\text{weight percent (w/w)} = \frac{\text{weight solute}}{\text{weight solution}} \times 100\%$$

$$\text{volume percent (v/v)} = \frac{\text{volume solute}}{\text{volume solution}} \times 100\%$$

$$\text{weight/volume percent (w/v)} = \frac{\text{weight solute, g}}{\text{volume solution, mL}} \times 100\%$$

Mass Percent (Mass %)

As the name suggests, the **mass percent** of any component in a solution is the mass of that component divided by the total mass of the solution times 100%:

$$\textbf{Mass percent} = \frac{\text{Mass of component}}{\text{Total mass of solution}} \times 100\%$$

$$\textbf{Parts per million (ppm)} = \frac{\text{Mass of component}}{\text{Total mass of solution}} \times 10^6$$

$$\textbf{Parts per billion (ppb)} = \frac{\text{Mass of component}}{\text{Total mass of solution}} \times 10^9$$

MOLARITY

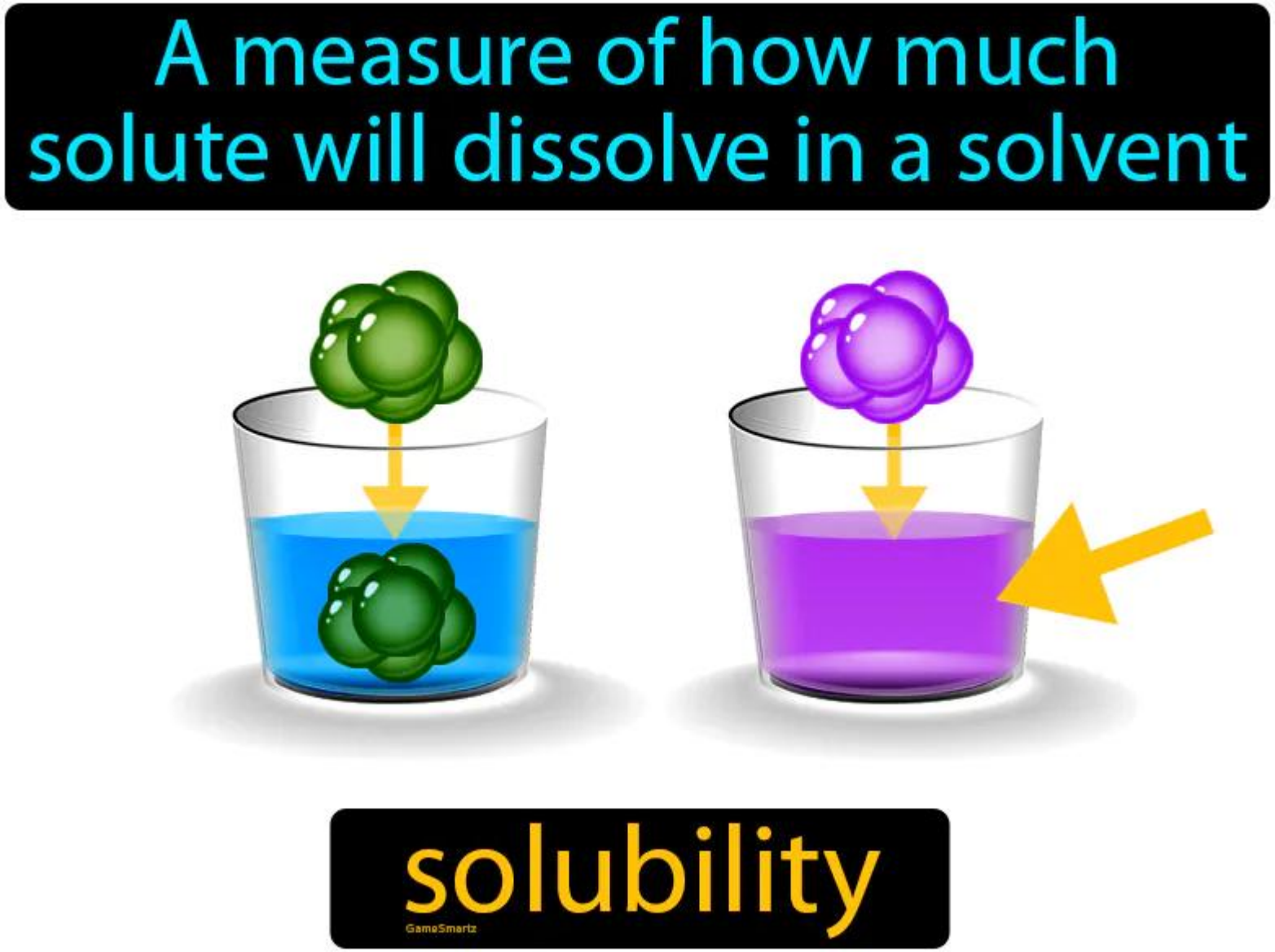
Number of moles of solute in a litre of solution.

$$\text{Molarity} = \frac{\text{moles solute}}{\text{1 L of solution}}$$

OR

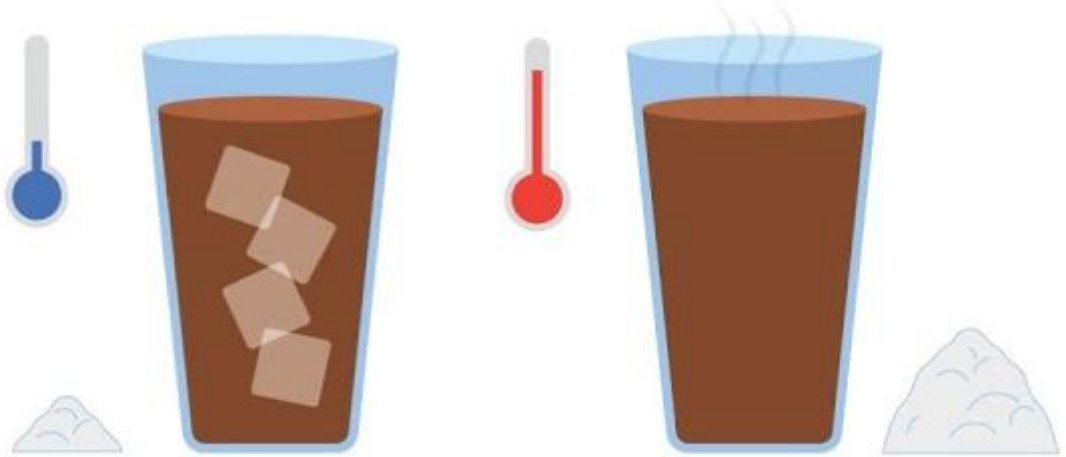
$$M = \frac{\text{mol}}{\text{L}}$$

1.5. Factors affecting solubility

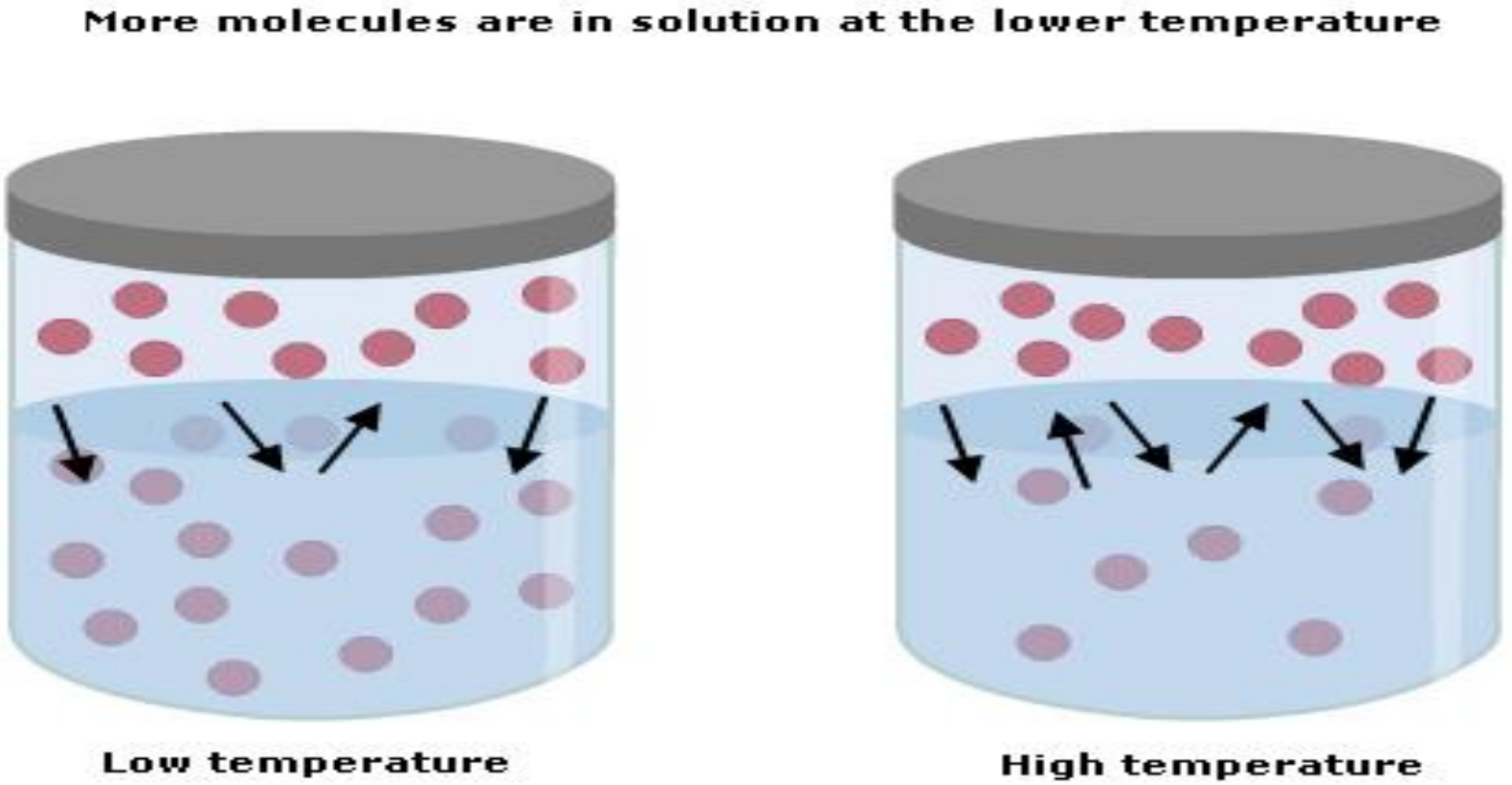


1.5. Factors affecting solubility

A. Effect of temperature on solubility

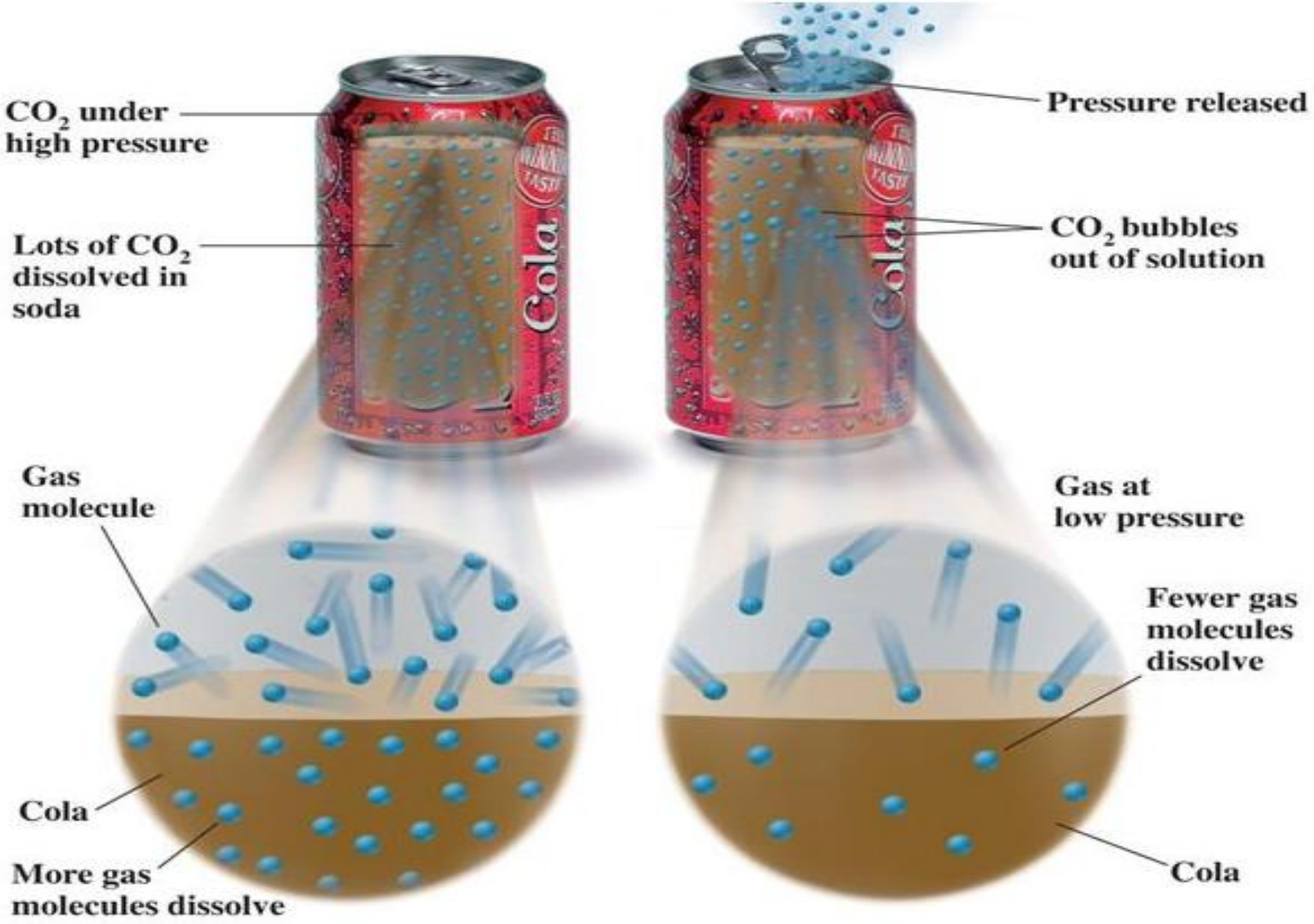


- The solubility of a **solid** in a liquid is significantly affected by temperature change.
- Solubility **increases** with **rising** temperature.



The temperature is **inversely proportional** to the solubility of the gas in a liquid, hence for all gases, as the **temperature increases**, **solubility decreases.**

B. Effect of Pressure on Solubility



Pressure has practically **no effect** on the solubility of **liquids** and **solids** but **affects the solubility of gases.**

According to **Henry's law**, the solubility of a gas in a liquid at a given temperature is directly proportional to the partial pressure of the gas over the solution:



Henry's law $\text{Solubility} = k \cdot P$

Solved Example

The Henry's-law constant of methyl bromide (CH_3Br), a gas used as a soil fumigating agent, is $k = 0.159\ \text{mol/(L} \cdot \text{atm)}$ at 25 °C. What is the solubility in mol/L of methyl bromide in water at 25 °C and a partial pressure of 125 mm Hg?

Answer



Henry's law $\quad \text{Solubility} = k \cdot P$

$k = 0.159\ \text{mol/(L} \cdot \text{atm)}$

$$P = 125\ \text{mm Hg} \times \frac{1\ \text{atm}}{760\ \text{mm Hg}} = 0.164\ \text{atm}$$

$$\text{Solubility} = k \cdot P = 0.159\ \frac{\text{mol}}{\text{L} \cdot \text{atm}} \times 0.164\ \text{atm} = 0.0261\ \text{M}$$